Lab 2: Requirements & Specification

Knowledge Is Power (KIP)

Hasibullah Noor
CS 411
Instructor: Thomas J. Kennedy
Individual Sections 1 & 2
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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements, overall behavior, and intended operation of the Knowledge Is Power (KIP) prototype. This document provides a common reference for the development team, instructors, testers, and other project stakeholders by describing the system scope, major functions, user characteristics, constraints, and functional requirements. The SRS is intended to guide implementation and evaluation of the prototype throughout CS 411.

1.2 Scope

Knowledge Is Power (KIP) is a web-based legal literacy platform designed to help young adults better understand everyday laws, legal responsibilities, and the possible consequences of their decisions. Young people increasingly encounter legal situations involving driving, digital conduct, employment, school, ownership, contracts, and personal safety while receiving limited practical instruction about how those laws apply in real situations. KIP addresses this problem through interactive, game-based learning that allows users to practice legal decision-making in a low-stakes environment.

The project is intended to make practical legal knowledge more accessible, understandable, and easier to retain. KIP uses clear explanations, realistic scenarios, immediate feedback, progress tracking, and source-backed legal content to help users recognize legally significant situations and make more informed decisions. The prototype demonstrates these goals through user accounts, a game browser, interactive scenarios, progress tracking, content recommendations, and legal source references.

1.3 Definitions, Acronyms, and Abbreviations

CDC: Centers for Disease Control and Prevention

KIP: Knowledge Is Power

MFCD: Major Functional Component Diagram

UX: User Experience

1.4 References

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[4] Touro Law Jacob D. Fuchsberg Law Center. (2017). A Lawyer's Experience in K-12 Law-Related Education: Lessons and Opportunities.

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1.5 Overview

The remainder of this SRS is organized into two major sections. Section 2 provides an overall description of KIP, including the product perspective, major product functions, user characteristics, constraints, and assumptions or dependencies. Section 3, developed collaboratively by the team, defines the detailed system features and functional requirements that guide implementation of the prototype.

2. Overall Description

2.1 Product Perspective

KIP is a web-based educational service built around learning modules, interactive games, user progress data, and source-backed legal content. The product is designed as a supplemental educational resource rather than a replacement for formal legal education or professional legal advice. Users interact with KIP through a web interface that provides access to accounts, games, scenarios, feedback, progress information, and recommendations.

The system is organized around presentation, application, and data functions. The presentation layer provides the interfaces that users directly interact with, including account access, the game browser, and scenario gameplay. The application layer manages authentication, scenario delivery, feedback, progress tracking, content curation, and legal source referencing. The data layer stores user, progress, game, topic, and legal-content information needed by the system. Together, these components support repeated, low-stakes practice and personalized reinforcement of legal concepts.

2.2 Product Functions

The KIP prototype provides several major software functions. The Authentication System manages user access and account sessions. The Module/Game Browser allows users to browse available learning content and games. The Scenario Engine and Scenario Player deliver interactive legal situations and response choices. The Feedback System explains the result of user decisions, while the Progress Tracker records completion and performance. The Curation System uses progress information to prioritize content for additional practice, and the Source Reference System connects instructional content to supporting legal sources.

The team's Real-World Product versus Prototype table identifies how the prototype narrows the larger product vision into a manageable implementation. The planned prototype focuses on Virginia traffic law, selected copyright-law use cases, staged game difficulty, and user account/history tracking.

The real-world version of KIP is intended to provide broader functionality than the prototype. The real-world product would provide traffic law coverage for federal law and all fifty states, comprehensive copyright law content, adaptive game difficulty with Relaxed, Standard, and Intense modes, and server-side storage for user accounts and history. The prototype reduces this scope to

Virginia traffic law, copyright law focused on social media platforms, three planned difficulty levels, and a database for storing user account and history information.

2.3 User Characteristics

KIP is primarily designed for young adults who are learning how everyday laws apply to situations they may encounter as they gain independence. These users are expected to be comfortable using common web applications but are not expected to have prior legal training. The interface therefore emphasizes clear language, concise explanations, and interactive practice rather than dense legal terminology.

Guest users may access selected game content with limited functionality, while registered users can use account-based features such as saved progress, history, and personalized recommendations. KIP is also intended to serve as a supplemental educational resource for parents, teachers, and school administrators. Administrative users may review content and legal-source information when material requires revision or becomes outdated.

2.4 Constraints

KIP is designed as a browser-based web application and is not intended to require downloads, installations, special peripherals, or advanced hardware beyond a modern web browser and internet connection. The project scope is limited to federal and Virginia law rather than exhaustive coverage of all jurisdictions or all areas of law. Because laws and policies can change, legal content must be reviewed and maintained so that instructional material and source references do not become obsolete.